Hotel Booking System

Report 2: UML Class Diagram Report
UML Class Diagram Modelling Assignment | Instructor: Eric Maniraguha
Kigali Independent University (ULK) | Submitted: October 2026
Group members: Maysara Dafaallh (202512952) | Ahmed Alhadi (202513597)

1. The UML Class Diagram

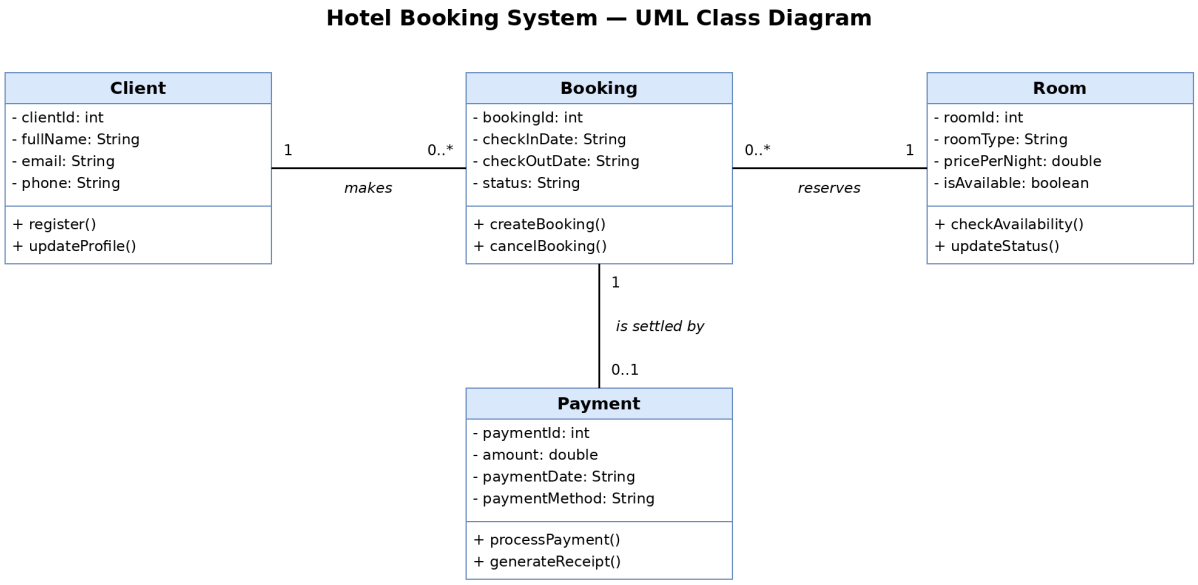


Figure 1: UML class diagram of the Hotel Booking System. In each class, the first compartment is the name, the second the attributes (private, marked -) and the third the methods (public, marked +).

2. Classes, Attributes and Methods

Class	Attributes	Methods
Client	- clientId: int - fullName: String - email: String - phone: String	+ register() + updateProfile()
Room	- roomId: int - roomType: String - pricePerNight: double - isAvailable: boolean	+ checkAvailability() + updateStatus()
Booking	- bookingId: int - checkInDate: String - checkOutDate: String - status: String	+ createBooking() + cancelBooking()
Payment	- paymentId: int - amount: double - paymentDate: String - paymentMethod: String	+ processPayment() + generateReceipt()

3. Relationships and Multiplicities

Association	Name	Multiplicity (end 1)	Multiplicity (end 2)	Meaning
Client – Booking	makes	Client: 1	Booking: 0..*	One client makes zero or many bookings; each booking is made by exactly one client.
Booking – Room	reserves	Booking: 0..*	Room: 1	One room can be reserved by zero or many bookings over time; each booking reserves exactly one room.
Booking – Payment	is settled by	Booking: 1	Payment: 0..1	A booking has at most one payment (it may not be paid yet); each payment belongs to exactly one booking.

4. Explanation of the Relationships

All three relationships are **simple associations**, drawn as plain lines with no arrowheads. No inheritance, aggregation or composition is used because the model does not require it: the four classes are independent concepts that only refer to each other.

Client – Booking (1 to 0..*): The client is the person who makes reservations. A client may register without booking yet (zero) or book many times, while every booking must be made by one identified client.

Booking – Room (0..* to 1): A booking reserves one room. The same room can appear in many bookings over time, but not for overlapping dates; a newly added room may have no booking yet.

Booking – Payment (1 to 0..1): The booking is the central class: it connects the client, the room and the payment. A booking may be created before it is paid, so the payment end is optional (0..1), but a payment always belongs to exactly one booking.

5. Assumptions Used for the Multiplicities

- Each booking is for one room only (no multi-room reservations).
- Each booking is paid by one single payment (no deposits or instalments).
- Bookings of the same room do not overlap in dates.
- Attribute types follow the project specification; dates are kept as String as required.

6. Notation Summary

- Three-compartment class boxes: name, attributes, methods.
- Visibility: - private (attributes), + public (methods).
- Multiplicity is written at each end of every association; the name of the association is written in italics under the line.